

Fine-tuning leaves a recoverable family history in model weights.

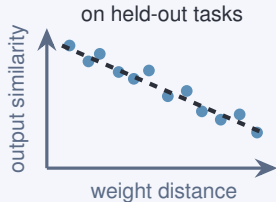
A. Can weights recover the tree?



✓ every true branch recovered in 85–100% of runs

(3% for a random tree); one attention key matrix,
0.1–0.2% of the weights, suffices

B. Do weights track behavior?



✓ closer weights, closer outputs

significant on every held-out
task for Llama; mixed for Flan

C. When does recovery work?



✓ label-free: no true tree needed

recovery odds $\times 5.9$ per
+0.1 in tree-likeness A